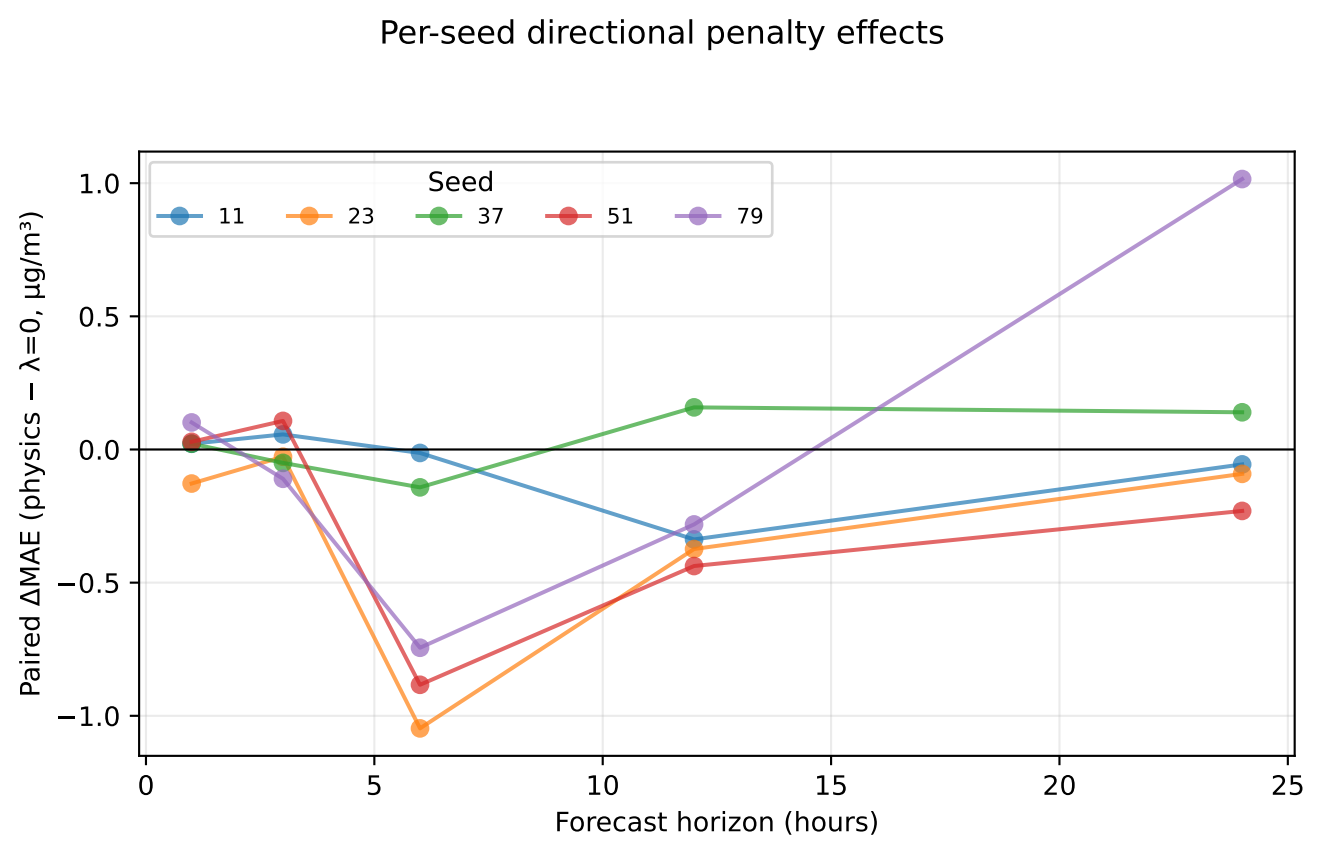
Per-seed directional penalty effects
Seed
11
23
37
51
79
Paired ΔMAE (physics − λ=0, μg/m³)
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
Forecast horizon (hours)